A Modular Base-e Parameterization of Lepton and Hadron Mass Ratios

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May 2026

Abstract

We present a compact phenomenological parameterization of selected lepton and hadron mass ratios using the constants e, π, the inverse fine-structure constant α^{-1}, the topological index $N = 5$, the bosonic-string reference dimension $D = 26$, and the Fibonacci number $F_{26} = 121393$. The construction is not derived from a Lagrangian and is not proposed as a replacement for the Standard Model. Its purpose is narrower: to record a low-complexity set of algebraic formulae, their numerical residuals, and the places where the ansatz fails or requires a non-standard interpretation.

1 Discrete Basis

The working basis is

$$P = \{2, 3, 5, 13, 137\}, \qquad N = 5, \qquad D = 26, \qquad F_{26} = 121393. \tag{1}$$

The integer 137 is treated as a discrete anchor. In the exploratory base-e notation used in this project,

$$137_e = 21100.0210102\ldots_e\,, \tag{2}$$

with the first fractional digit pattern

$$d_{-1}, d_{-2}, d_{-3}, d_{-4}, d_{-5}, d_{-6}, d_{-7} = 0, 2, 1, 0, 1, 0, 2. \tag{3}$$

The earlier digit moment

$$I_5 = \sum_{k=1}^{5} k^2 d_{-k} = 42 \tag{4}$$

has the equivalent topological form

$$I_5 = 2(D - N) = 2(26 - 5) = 42. \tag{5}$$

The core compression operator is

$$q = \frac{N(N-2)}{e^4\pi^3} = \frac{15}{e^4\pi^3} = 0.008860611874290873. \tag{6}$$

We also use

$$s = \frac{\alpha}{2\pi}, \qquad \delta\phi = \cos(1/N) - \cos(2/N) = 0.05900558383835652. \tag{7}$$

2 Charged Lepton Sector

The electron is used as the normalization anchor. The project-level normalization is

$$C_e = \frac{1}{1 - s\left(N + \frac{2}{eN}\right)}, \tag{8}$$

and the electron scale is written as

$$m_e = 10e^{11}\cos^2(2/N)C_e. \tag{9}$$

This equation is a normalization convention in the present note.

The muon ratio is

$$\frac{m_\mu}{m_e} = \alpha^{-1}\left(\frac{3}{2} + q\right). \tag{10}$$

The tau ratio is

$$\frac{m_\tau}{m_e} = \alpha^{-1}\left[N^2 + qf_\tau\right], \tag{11}$$

where

$$f_\tau = 2(D - N) + 2e^{-2} + 2e^{-7}. \tag{12}$$

3 Hadron and Baryon Benchmarks

For the neutral pion, the baseline expression is

$$\frac{m_{\pi^0}}{m_e} = \alpha^{-1}\left[2 - \delta\phi - q\left(1 + \frac{1}{N} + \frac{1}{\pi}\right)\right]. \tag{13}$$

The Phase 9 audit showed that the literal replacement $q \to 117/137$ fails by changing the scale of the expression. The retained compact modulation uses the invariant

$$117 = DN - 13 = 9 \cdot 13 \tag{14}$$

only as a small correction:

$$q_{117} = q\left(1 - \frac{2^2}{117 \cdot 137}\right). \tag{15}$$

The accepted pion benchmark is therefore

$$\frac{m_{\pi^0}}{m_e} = \alpha^{-1}\left[2 - \delta\phi - q_{117}\left(1 + \frac{1}{N} + \frac{1}{\pi}\right)\right]. \tag{16}$$

For the proton ratio, the low-complexity Phase 9 expression is

$$K_p = \frac{M_p}{m_e} = \frac{2F_{26}}{DN}\cos(1/N)\left[1 + es + \frac{6}{137^2(D + N + 2^2)}\right]. \tag{17}$$

The integer 6 is the Euler-phi gap

$$\phi(137) - DN = 136 - 130 = 6. \tag{18}$$

This should be read as a compact empirical candidate, not as a derivation from QCD.

4 Predicted Sterile Benchmarks

The charged heavy-lepton extrapolation is

$$\frac{m_{\tau'}}{m_e} = \alpha^{-1}\left[D^2 + q(DN)\right], \tag{19}$$

which gives

$$m_{\tau'} = 47.417730830364825 \text{ GeV}. \tag{20}$$

This is above the project-level 45 GeV reference threshold, but below the conventional sequential charged-heavy-lepton limit of about 100.8 GeV. It therefore cannot be interpreted as an ordinary Standard-Model-like fourth charged lepton without additional assumptions.

The sterile neutral benchmark is

$$m_{\nu_{\tau'}} = m_e \left(\frac{F_{26}}{\alpha^{-1}}\right)\left[1 + \frac{DN}{\pi}\delta\phi(1-s)\right], \tag{21}$$

which gives

$$m_{\nu_{\tau'}} = 1.5566463427697075 \text{ GeV}. \tag{22}$$

This mass is kinematically cold, but the relic abundance, lifetime, production history, and visible decay channels are not determined by the mass formula. The benchmark is conditionally viable only for a fully sterile or otherwise sufficiently decoupled state. A simple Fermi-LAT/Galactic-Center audit did not support a direct gamma-ray-excess interpretation: the phase-modulated annihilation estimates are far below the nominal $\sim 2 \times 10^{-26}$ $\text{cm}^3\,\text{s}^{-1}$ scale.

5 Numerical Summary

Table 1: Mass-ratio benchmarks recorded by the model. Residuals are quoted relative to the reference values used in the project audit.

Observable	Formula family	Model value	Residual
m_μ/m_e	$\alpha^{-1}(3/2+q)$	206.76822	-0.29 ppm
m_τ/m_e	$\alpha^{-1}(N^2+qf_\tau)$	3477.22820	-0.022 ppm
m_{π^0}	q_{117} pion branch	134.976797 MeV	-0.020995 ppm
M_p/m_e	$\phi(137)$ gap branch	1836.152490	-0.098658 ppm
$m_{\tau'}$	$\alpha^{-1}[D^2+qDN]$	47.417731 GeV	benchmark
$m_{\nu_{\tau'}}$	sterile branch	1.556646 GeV	benchmark

6 Conclusion

The compact relations collected here are best viewed as a modular parameterization of selected particle scales. The formulas are numerically tight and reproducible in the accompanying code, but they remain phenomenological until connected to a dynamical field theory, collider constraints, and standard hadronic calculations. The useful output of the exercise is therefore not a claim of replacement physics, but a falsifiable matrix of low-complexity numerical benchmarks.